

| Assessment | Test | Domain        | Skill                  | Difficulty                                          |
|------------|------|---------------|------------------------|-----------------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Medium <div><div></div><div></div><div></div></div> |

Question

$\sqrt[3]{x^3y^6}$

Which of the following expressions is equivalent to the expression above?

Answer

- A.  $y^2$
- B.  $xy^2$
- C.  $y^3$
- D.  $xy^3$

Correct Answer: B

Rationale

Choice B is correct. One of the properties of radicals is  $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$ . Thus, the given expression can be rewritten as  $\sqrt[3]{x^3} \cdot \sqrt[3]{y^6}$ .

Simplifying by taking the cube root of each part gives  $x^1 \cdot y^2$ , or  $xy^2$ .

Choices A, C, and D are incorrect and may be the result of incorrect application of the properties of exponents and radicals.